Day 20 | Wednesday

December 2, 2026

20

Angles, triangles and drawing interpretation

12:45 PM–2:00 PM • Math • 75 min

Pythagorean theorem and brace lengths

2:15 PM–3:30 PM • Math • 75 min

Week 6 • 2 scheduled blocks • 150 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

12:45 PM–2:00 PM • 75 min

Angles, triangles and drawing interpretation

12:45 PM–12:50 PM	**Start**	5
12:50 PM–12:55 PM	**Recall**	5
12:55 PM–1:10 PM	**Teach**	15
1:10 PM–1:25 PM	**Together**	15
1:25 PM–1:45 PM	**Practise**	20
1:45 PM–1:55 PM	**Check**	10
1:55 PM–2:00 PM	**Wrap up**	5

Math • Math Week 6, Lesson 5

2:15 PM–3:30 PM • 75 min

Pythagorean theorem and brace lengths

2:15 PM–2:20 PM	**Start**	5
2:20 PM–2:25 PM	**Recall**	5
2:25 PM–2:40 PM	**Teach**	15
2:40 PM–2:55 PM	**Together**	15
2:55 PM–3:15 PM	**Practise**	20
3:15 PM–3:25 PM	**Check**	10
3:25 PM–3:30 PM	**Wrap up**	5

Math • Math Week 6, Lesson 6

Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time

Angles, triangles and drawing interpretation

Today’s goal: Classify angles and use straight-line and triangle angle totals to find a missing angle.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 12.1–12.3 pp.188–190

Exercise Book 12A Q1–9 (pp.75–76); 12B Q2 (p.77); 12C Q3–4 (p.79)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 12.1–12.3 pp.188–190

Recall and reconnect

Try to identify 90°, a straight angle and a full turn; use Exercise Book 12A Q3.

Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 12A Q1–9 (pp.75–76); 12B Q2 (p.77); 12C Q3–4 (p.79)

Textbook sections 12.1–12.3 pp.188–190

Recall from the Exercise Book | 12A Q3

Name each angle: 12°, 90°, 179°, 95°.

Write the operation, working and units before checking.

Textbook sections 12.1–12.3 pp.188–190

Use the correct angle total

1 A right angle is 90°; a straight angle is 180°; a full turn is 360°.

2 Complementary angles add to 90°.

3 Supplementary angles add to 180°.

4 Mark the relevant total on the drawing before subtracting a known angle.

Textbook sections 12.1–12.3 pp.188–190

Find a missing angle in a triangle

1 The three interior angles of a triangle add to 180°.

2 Missing angle = 180° − the other two interior angles.

3 An adjacent outside angle is a different angle from the one inside.

4 Check that your three interior angles total 180°.

Textbook sections 12.1–12.3 pp.188–190

Match the missing angle to its total

Complementary angles • Total 90°.
If one is 33°, the other is 90° − 33° = 57°.

Supplementary angles • Total 180°.
If one is 128°, the other is 180° − 128° = 52°.

Triangle example • Known angles 90° and 52°.
Missing angle = 180° − 90° − 52° = 38°; it is acute.

Together • Work 12A Q4–8. Mark known angles and the relevant total.
Shape Canvas: vary one triangle angle and predict the consequence.
Numeracy Hub: inspect diagrams; advanced protractor topics are not required.

Classroom examples demonstrate the three relationships • Textbook pp.188–190 • Sketch and label before calculating

Textbook | page 190

12.3 Angles in a triangle

The three angles inside any triangle add to 180°. You can see why by tearing the three corners off a paper triangle and laying them side by side. They fit together into a straight line.



Figure 12.3 The corners of a 50°, 60°, 70° triangle (left) placed side by side make a straight angle: 50° + 60° + 70° = 180°.

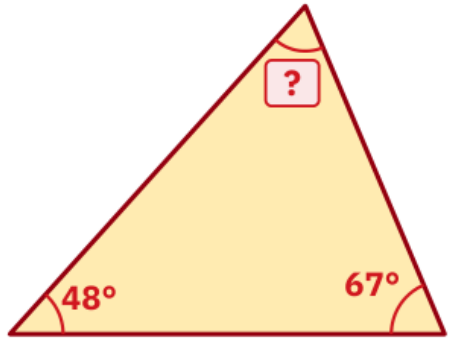
Textbook p.190 • Figure 12.3 — Triangle corners form a straight angle

Textbook sections 12.1–12.3 pp.188–190

Textbook | page 190

Example 12.3 The third angle of a gusset

A triangular gusset has angles of 48° and 67°. Find the third angle.



SOLUTION

1. The three angles add to 180°.
2. Third angle = 180° − 48° − 67° = 65°.

Answer 65°

Textbook p.190 • Example 12.3 — Find the missing gusset angle

Textbook sections 12.1–12.3 pp.188–190

Spot the mistake. Explain the correction.

A diagram shows an angle beside the triangle. Is it automatically an interior angle?

Discuss the method before checking the answer.

No. Identify the triangle’s interior first; use the straight-line relationship if the label is outside.

Textbook sections 12.1–12.3 pp.188–190

Work together

Solve Exercise Book 12A Q4–8 in pairs. Mark known angles and the relevant total on each sketch.

Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 12A Q1–9 (pp.75–76); 12B Q2 (p.77); 12C Q3–4 (p.79)

Textbook sections 12.1–12.3 pp.188–190

Guided practice | 12A Q4

Two angles sit on a straight line. One is 64°. Find the other.

Work with a partner. Take turns explaining the next step.

Textbook sections 12.1–12.3 pp.188–190

Compare your method | 12A Q4

Two angles sit on a straight line. One is 64°. Find the other.

CHECK TOGETHER

116°

Textbook sections 12.1–12.3 pp.188–190

Your scheduled practice

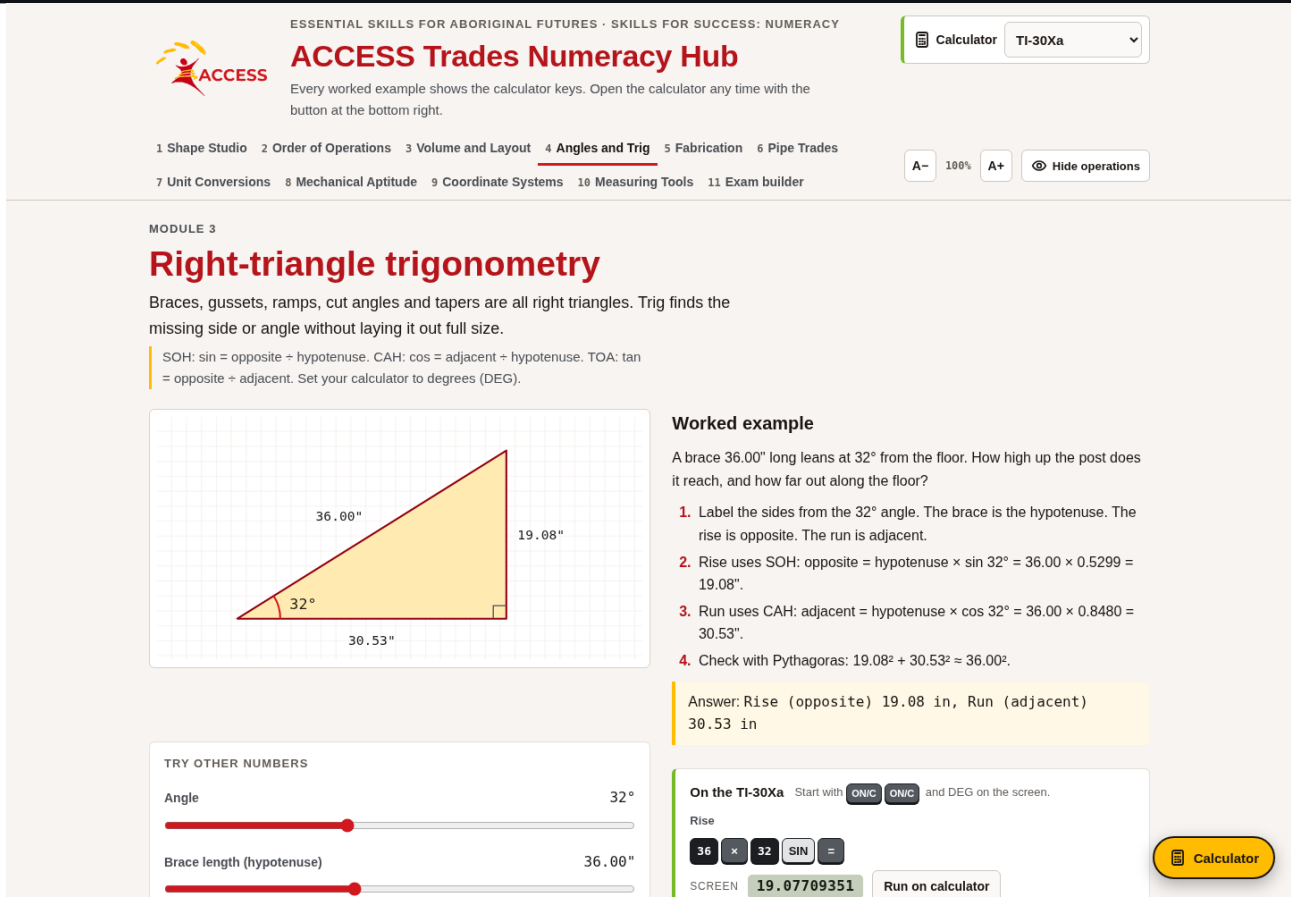
Complete Exercise Book 12A Q1–2,9 and 12B Q2; then 12C Q3–4. Use Shape Canvas to vary one angle and explain the consequence.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 12A Q1–9 (pp.75–76); 12B Q2 (p.77); 12C Q3–4 (p.79)

Textbook sections 12.1–12.3 pp.188–190

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook sections 12.1–12.3 pp.188–190

Practise in Numeracy Hub and Shape Canvas

1. Use a Shape Canvas triangle and identify its three interior angles.

2. Predict how changing one angle affects the remaining angle total, then compare the labels.

3. Inspect the corresponding diagrams in Numeracy Hub Angles and Trig; omit advanced protractor topics.

Textbook sections 12.1–12.3 pp.188–190

Practise from the Exercise Book | 12A Q1

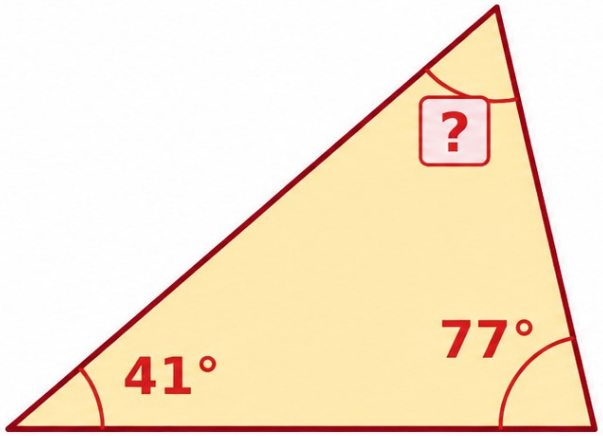
How many degrees are in one third of a turn?

Write the operation, working and units before checking.

Textbook sections 12.1–12.3 pp.188–190

Practise from the Exercise Book | 12C Q3

Find the third angle.



Write the operation, working and units before checking.

Textbook sections 12.1–12.3 pp.188–190

Check independently

Exercise Book 12A Q6–7 and Exercise Book 12B Q2.

Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 12A Q1–9 (pp.75–76); 12B Q2 (p.77); 12C Q3–4 (p.79)

Textbook sections 12.1–12.3 pp.188–190

Independent check | 12A Q6

A triangle has angles of 52° and 71°. Find the third angle.

Answer independently. Keep worked solutions closed.

Textbook sections 12.1–12.3 pp.188–190

Independent check | 12A Q7

One acute angle of a right triangle is 38°. Find the other.

Answer independently. Keep worked solutions closed.

Textbook sections 12.1–12.3 pp.188–190

Independent check | 12B Q2

A flat bar is bent so that one side of the bend makes 142° with the straight part. What angle is on the other side of the bend? Is it acute or obtuse?

Answer independently. Keep worked solutions closed.

Textbook sections 12.1–12.3 pp.188–190

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 12A Q1–9 (pp.75–76); 12B Q2 (p.77); 12C Q3–4 (p.79)

Textbook sections 12.1–12.3 pp.188–190

Break

2:00 PM–2:15 PM

Return ready for: Pythagorean theorem and brace lengths

This break is excluded from teaching minutes.

Pythagorean theorem and brace lengths

Today’s goal: Find a missing side of a right triangle and verify the result with the squared-side relationship.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Exercise Book 12A Q10–16 (p.76); 12B Q3 (p.77); 12C Q1–2 (p.79)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Recall and reconnect

Label the hypotenuse opposite the right angle. Retrieve a square and a square root.

Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 12A Q10–16 (p.76); 12B Q3 (p.77); 12C Q1–2 (p.79)

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Identify the hypotenuse first

1 Pythagoras applies to a right triangle.

2 The hypotenuse is opposite the 90° angle and is the longest side.

3 Write $a^2 + b^2 = c^2$ with c as the hypotenuse.

4 Use matching units before squaring side lengths.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Choose addition or subtraction

1 For the hypotenuse, $c = \sqrt{(a^2 + b^2)}$.

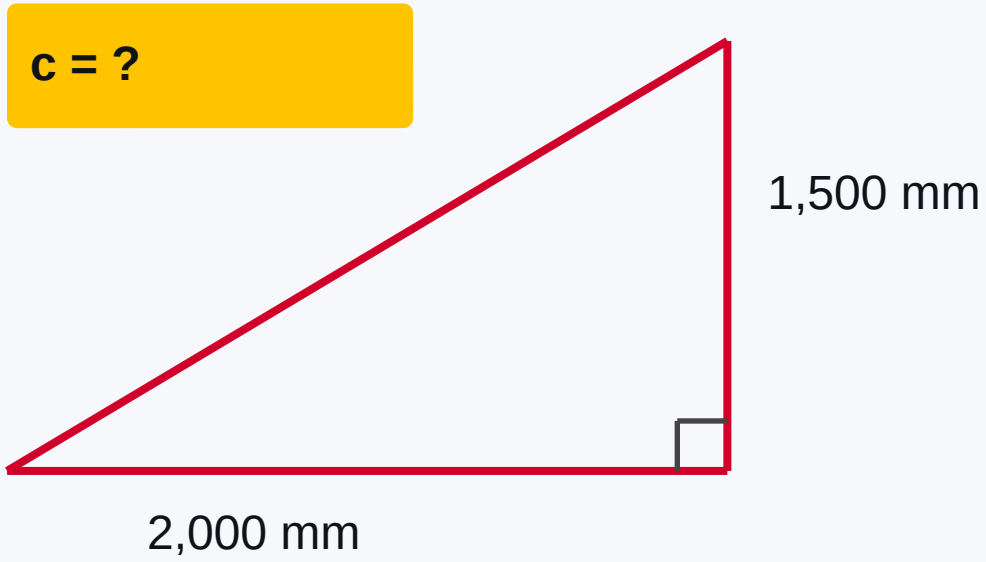
2 For a missing leg, subtract the known leg’s square from c^2 before taking the root.

3 Bracket the complete sum or difference under the square root.

4 Check the squared relationship and whether the hypotenuse is longest.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Square, combine, then take the root



$c = \sqrt{(1{,}500^2 + 2{,}000^2)}$
$= \sqrt{6{,}250{,}000}$
$= 2{,}500$ mm

Check: $2{,}500^2 = 6{,}250{,}000$.
The answer exceeds both legs.
For a missing leg, subtract squares.

Together • Work 12A Q10–13. Draw and label every triangle first.
Explain the bracketed quantity before entering the square-root key.

Classroom right-triangle example • Diagram is schematic • Use the selected calculator’s root-entry sequence

Textbook | page 192

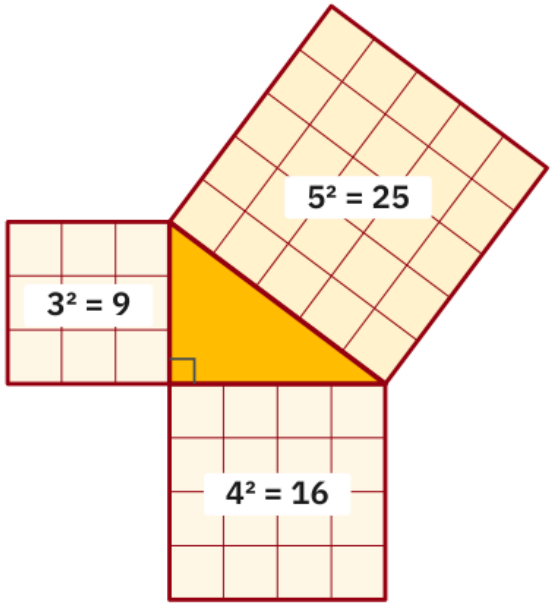


Figure 12.5 The squares on the sides of a 3-4-5 right triangle: 9 + 16 = 25.

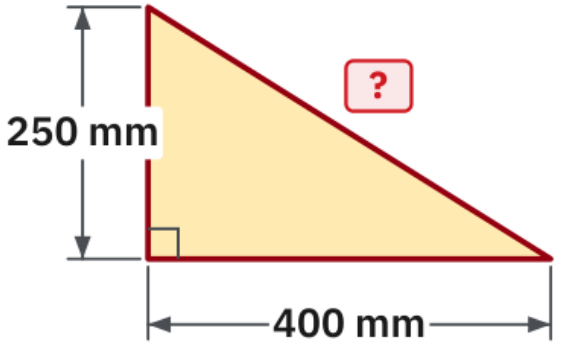
Textbook p.192 • Figure 12.5 — Squares on a 3–4–5 triangle

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Textbook | page 192

Example 12.4 Length of a gusset edge

A triangular gusset has legs of 400 mm and 250 mm. Find the length of the sloped edge (the hypotenuse) to two decimal places.



SOLUTION

1. $c^2 = 400^2 + 250^2 = 160{,}000 + 62{,}500 = 222{,}500.$
2. $c = \sqrt{222{,}500}.$

Textbook p.192 • Example 12.4 — Find a gusset hypotenuse

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Spot the mistake. Explain the correction.

A missing leg is calculated by adding the two known squares. What should change?

Discuss the method before checking the answer.

Use the hypotenuse squared minus the known leg squared, then take the square root.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Work together

Solve Exercise Book 12A Q10–13; learners draw and label each triangle before entering keys.

Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 12A Q10–16 (p.76); 12B Q3 (p.77); 12C Q1–2 (p.79)

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Guided practice | 12A Q10

Legs of 700 mm and 2,400 mm. Find the hypotenuse.

Work with a partner. Take turns explaining the next step.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Compare your method | 12A Q10

Legs of 700 mm and 2,400 mm. Find the hypotenuse.

CHECK TOGETHER

2,500 mm

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Your scheduled practice

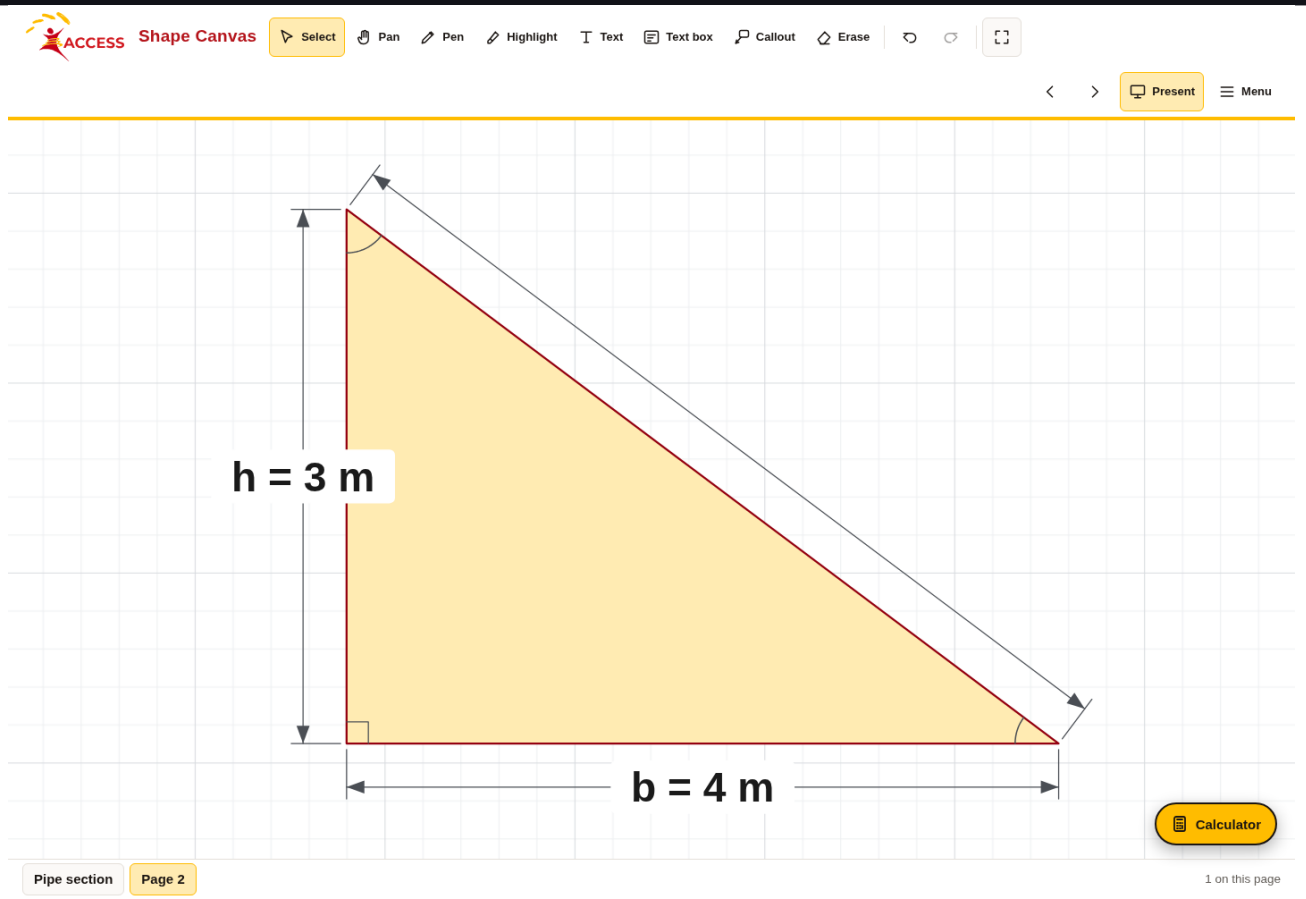
Complete Exercise Book 12A Q15–16, 12B Q3 and 12C Q1–2. Check Shape Canvas answers only after writing the equation.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 12A Q10–16 (p.76); 12B Q3 (p.77); 12C Q1–2 (p.79)

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Practise in Numeracy Hub and Shape Canvas

1. Create a right triangle in Shape Canvas and hide the hypotenuse or one leg.

2. Draw the same triangle and write the correct squared-side equation before calculating.

3. Use Numeracy Hub Calculator for the bracketed sum or difference; reveal the Canvas answer last.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Practise from the Exercise Book | 12A Q15

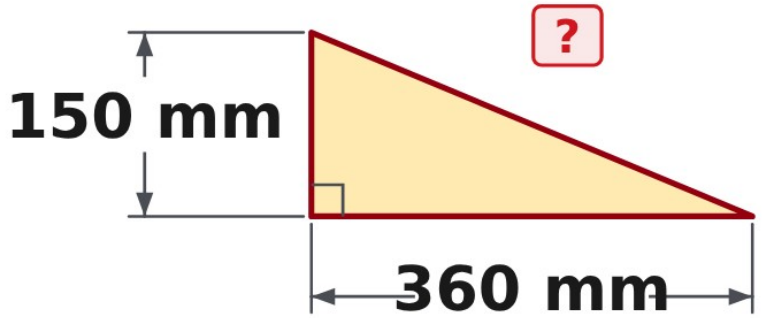
Do 8, 15 and 17 make a right triangle? Show the check.

Write the operation, working and units before checking.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Practise from the Exercise Book | 12C Q1

Find the hypotenuse.



Write the operation, working and units before checking.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Check independently

Exercise Book 12A Q10 and Q12 with the right-angle diagram and units.

Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 12A Q10–16 (p.76); 12B Q3 (p.77); 12C Q1–2 (p.79)

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Independent check | 12A Q10

Legs of 700 mm and 2,400 mm. Find the hypotenuse.

Answer independently. Keep worked solutions closed.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Independent check | 12A Q12

Hypotenuse 1,700 mm, one leg 800 mm. Find the other leg.

Answer independently. Keep worked solutions closed.

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 12A Q10–16 (p.76); 12B Q3 (p.77); 12C Q1–2 (p.79)

Textbook sections 12.4 pp.191–193; calculator roots pp.31–33